

Chapter 19: Life Cycle of Stars

Textbook Exercise Solutions & Answers



1. Search and you will find

(a) Our galaxy is called

Our galaxy is called the Milky Way and Mandakini.

(b) For measuring large distances, is used as a unit.

For measuring large distances, the light year is used as a unit.

(c) The speed of light is km/s.

The speed of light is 300000 km/s.

(d) There are about stars in our galaxy.

There are about 100 billion stars in our galaxy.

(e) The end stage of the Sun will be

The end stage of the Sun will be a white dwarf.

(f) Stars are born out of clouds.

Stars are born out of interstellar clouds.

(g) Milky way is a galaxy.

Milky way is a spiral galaxy.

(h) Stars are gigantic spheres of gas.

Stars are gigantic spheres of hot gas.

(i) The masses of other stars are measured relative to the mass of the

The masses of other stars are measured relative to the mass of the Sun.

(j) Light takes to reach us from the Sun while it takes to reach us from the moon.

Light takes about 8 minutes to reach us from the Sun while it takes about 1 second to reach us from the moon.

(k) The larger the mass of a star, the faster is its

The larger the mass of a star, the faster is its evolution.

(l) The number of fuels used in the life of a star depends on its

The number of fuels used in the life of a star depends on its mass.



The figure shows a spiral galaxy. Our solar system is situated in a similar galaxy.

2. Who is telling lies?

(a) Light year is used to measure time.

False. (Light year is used to measure distance.)

(b) End stage of a star depends on its initial mass.

True.

(c) A star ends its life as a neutron star when the pressure of its electrons balances its gravity.

False. (A star ends its life as a neutron star when the pressure of its neutrons balances its gravity.)

(d) Only light can emit from the black hole.

False. (Not even light can be emitted by a black hole.)

(e) The Sun will pass through the supergiant stage during its evolution.

False. (The Sun will pass through the red giant stage during its evolution.)

(f) The Sun will end its life as a white dwarf.

True.

3. Answer the following question

(a) How do stars form?

There are huge clouds of gas and dust in the empty spaces between the stars in a galaxy. These clouds are called interstellar clouds. The size of an interstellar cloud is about a few light years.



A picture of interstellar clouds taken by the Hubble space telescope

When an interstellar cloud starts contracting due to some disturbance, its density and temperature increase. This results in formation of a dense sphere of hot gas and nuclear energy generation starts at the centre of the star. Therefore, the gas sphere becomes self-luminous. Thus a star is formed, i.e., a star is born.

A huge interstellar cloud can produce thousands of stars at a time.

(b) Why do stars evolve?

Although stars appear stable for quite a long period of time, their properties do change, though very slowly. A change in the properties of a star, leading to its passing through different stages, is called evolution of the star. Burning of the fuel at the centre of the star and a gradual decrease in its amount is the main reason of evolution of a star.

(c) What are the three end stages of stars?

1. Stars having initial mass less than 8 times the mass of the Sun ultimately become white dwarfs.
2. Stars having initial mass between 8 and 25 times the mass of the Sun ultimately become neutron stars.
3. Stars having initial mass larger than 25 times the mass of the Sun ultimately turn into black holes.

(d) Why was the name black hole given?

When a star having initial mass larger than 25 times the mass of the Sun reaches its end stage, its gravitational force and density increase exponentially. All nearby objects get attracted towards the star and nothing can come out of it, not even light. All incident light is absorbed by the star. We can probably see a very minute black hole at its place. Hence, the name is given as black hole.

(e) Which types of stars end their life as a neutron star?

Stars having initial mass between 8 and 25 times the mass of the Sun end up as neutron stars. When these stars pass through the supergiant stage, their size increases to 1000 times. Huge explosion that occurs in the last stage of these stars is very powerful and very high energy is given off. After the huge explosion, called the supernova explosion, their central portion contracts in size to about 10 km. In this stage, such stars are completely made up of neutrons and hence are called neutron stars.

4. Write about the following in your own words

(a) If you are the Sun, write about your properties in your own words.

There are billions of stars in the galaxy called the Milky Way and Mandakini. I am one of the small stars called the Sun. I have my own family called the solar system. There are planets, satellites, asteroids, comets and meteors in my family. My mass is 2×10^{30} kg which is about 3.3 lakh times that of the earth. With a radius of 695700 km, my size is about 100 times that of the earth. My surface temperature is about 5800 K while it is 1.5×10^7 K at the centre.

72 % of my mass consists of hydrogen while 26 % consists of helium. Rest 2 % is made up of elements heavier than helium. I am about 4.5 billion years old and the scientists on the earth have concluded that not much change has taken place in my properties during this period. According to the scientists, in the end stage of my life, I will turn into a red giant star. Thereafter, I will first explode and then contract to become as small as the earth. I will appear small as well as white and hence, I will be called a white dwarf. This will be my last stage for ever.

(b) Describe white dwarfs.

Depending on the initial mass, stars can reach one of the three end stages. White dwarf is one such stage of stars having initial mass less than 8 times the mass of the Sun. These stars undergo huge expansion and their radius increases by a factor of 100 to 200. These stars appear reddish because of their large size and lower temperature. Hence, these stars are also called red giant stars.

At the end of their evolution, these stars explode, their outer gas envelope is thrown outward and the inner part contracts to the size of the earth. However, the density in the star becomes very high. In this stage, the pressure due to electrons becomes independent of temperature and sufficient to balance the gravitational force for ever.

In this stage, such stars look white and due to their small size they are called white dwarfs.